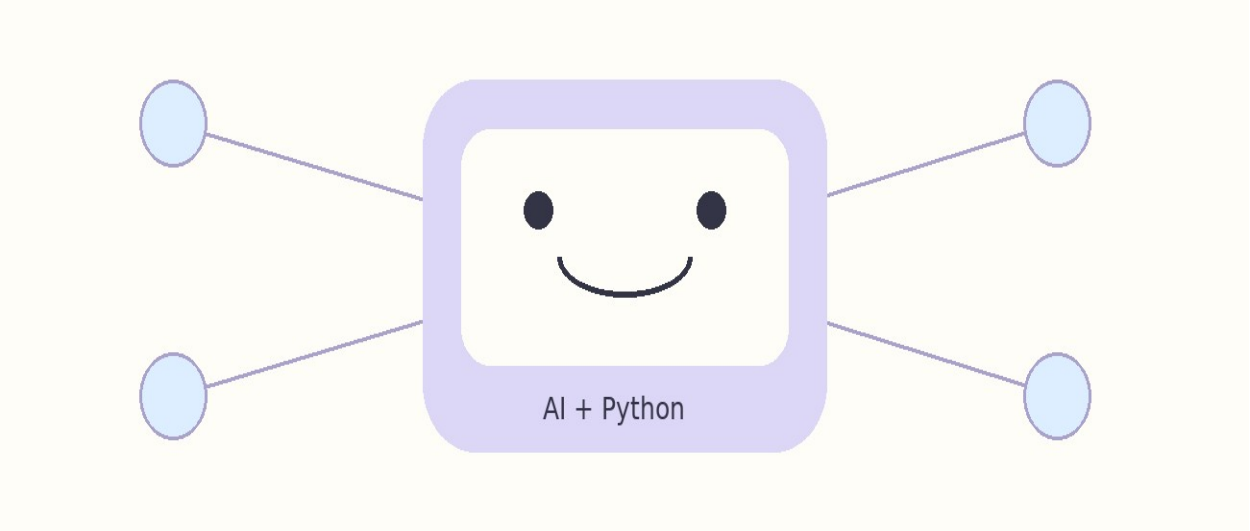
WEEK 1 ASSIGNMENT

INTRODUCTION TO ARTIFICIAL INTELLIGENCE & PYTHON



Foundations of intelligent systems and practical Python programming

Submission detail	Information
Internship organization	InternNova
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Introduction to Artificial Intelligence

Task 1 • 20 marks | Concepts and system types

1.1 What is Artificial Intelligence?

AI is the study and engineering of computer systems that perform tasks associated with intelligence, including reasoning, learning, perception, language and planning. An AI system can follow explicitly encoded rules, learn patterns from examples, or combine both approaches. A useful definition focuses on capabilities and behavior rather than assuming human consciousness.

Main objectives

The main objectives are to automate selected tasks, infer useful information from data, support decisions, communicate through language and select actions that help achieve a goal. A medical imaging assistant, for instance, may flag an unusual scan for a clinician; it does not need to possess general human intelligence to be useful.

How an AI system works

A typical workflow begins with a task and an input, such as an image, a question or a set of measurements. Input is prepared, a rule set or learned model processes it, and the system produces a prediction, recommendation, response or action. The result is evaluated against the intended goal. Monitoring and human review are important because input conditions and error consequences vary.

INPUT → PREPARATION → RULES / MODEL → OUTPUT → EVALUATION
Example: leaf photograph → image preprocessing → classifier → suspected disease label → expert review.

1.2 Types of AI by capability

Type	Meaning and example	Development status
ANI — Artificial Narrow Intelligence	Specialized capabilities, e.g., spam filtering, recommendation or language assistance. A system can support many tasks without proving human-level generality.	Deployed today; performance is bounded by task, data and design.
AGI — Artificial General Intelligence	A proposed system with broad, adaptable intelligence across domains at a human-comparable level. A hypothetical agent that reliably learns unfamiliar intellectual work illustrates the idea.	Not established as achieved under a generally accepted standard; definitions and evaluations remain debated.
ASI — Artificial Superintelligence	A hypothetical intelligence exceeding humans broadly, not merely on one game or calculation.	Theoretical; not established as achieved.

AGI and ASI examples above are hypothetical, not claims about existing products. Strong performance on a benchmark alone does not establish either category.

AI, Machine Learning & Deep Learning

1.3 Understanding the relationship

AI is the broad field. Machine learning (ML) is a subset of AI in which models learn patterns from data instead of requiring a programmer to specify every decision rule. Deep learning (DL) is a subset of ML based on neural networks with multiple processing layers. Not all AI uses ML, and not all ML uses deep neural networks.

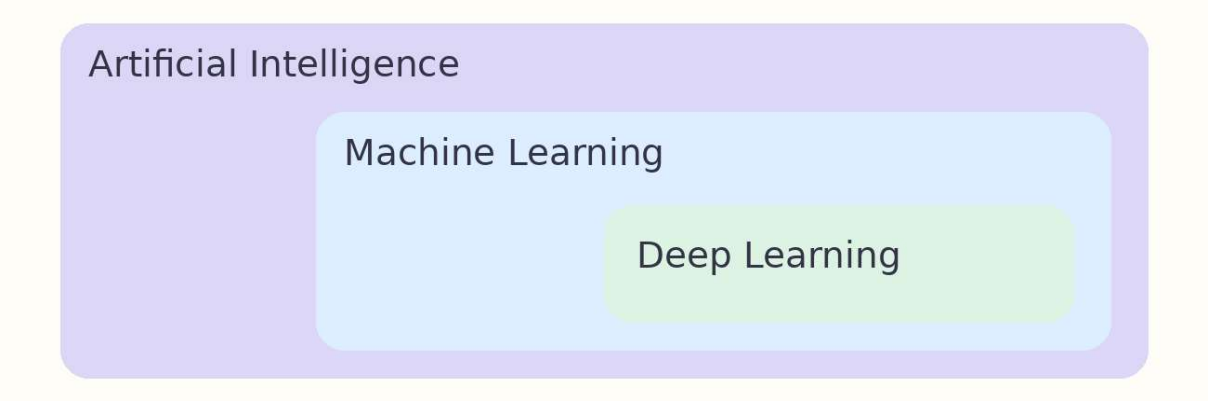


Diagram 1. Conceptual hierarchy: Deep Learning ⊂ Machine Learning ⊂ Artificial Intelligence.

Aspect	Artificial Intelligence	Machine Learning	Deep Learning
Scope	Broad intelligent behavior	Learning patterns from data	Learning with multilayer neural networks
Typical approach	Rules, search, planning or learning	Statistical fitting and optimization	Layered representations and optimization
Example	Rule-based expert system	Spam classifier trained on labeled messages	Neural image classifier
Data requirement	May rely on hand-written knowledge	Depends on task and method	Often substantial for training from scratch
Compute and explanation	Varies; rules may be inspectable	Varies by algorithm	Often compute-intensive; explanation may be harder

A practical distinction

A programmer can write “if marks ≥ 40, return Pass.” This is a hand-coded decision rule: it does not learn from previous students. An ML system might instead learn a relationship from labeled records. A DL model would learn using a multilayer network, usually unnecessary for such a simple threshold task.

Do not equate complexity with usefulness. Choose the simplest approach that solves the task safely and reliably. A transparent rule can be preferable when the policy is already known.

Applications, Advantages & Limitations

1.4 Real-life application areas

Area	Practical use	Human oversight / caution
Healthcare	Image analysis can flag suspicious regions in scans.	Clinicians validate findings; errors affect care.
Education	Adaptive practice can recommend exercises.	Teachers review quality and accessibility.
Banking	Fraud detection flags unusual transactions.	Analysts review false alarms and fairness.
Transportation	Perception systems identify road objects.	Safety validation and operating limits matter.
Entertainment	Recommendations rank films or music.	Preferences and filter effects need attention.
Virtual assistants	Speech and language systems answer requests.	Privacy and misunderstood requests matter.
Agriculture	Image models identify possible crop disease.	Farmers verify under local field conditions.

1.5 Advantages

Automation reduces repetitive manual effort. Speed enables rapid processing of large datasets. Accuracy can improve on suitable, well-defined tasks when systems are properly trained and validated. Availability supports continuous service, subject to infrastructure uptime. Data analysis can reveal patterns that help people prioritize investigation. Benefits depend on the task and how performance is measured.

1.6 Limitations and responsible use

Bias can arise from unrepresentative data, labels or design decisions. Privacy concerns arise when sensitive information is collected or exposed. Development costs include data preparation, computing, testing and maintenance. Dependence on data can cause failures when new situations differ from training examples. Systems lack dependable human judgment about context, values and responsibility. Automation may also displace or change jobs, creating a need for reskilling and organizational planning.

Responsible practice: define the task; minimize sensitive data; test for errors and unequal outcomes; provide human review; monitor changes; and document limitations. High confidence is not the same as correctness.